StockPilot

Inventory tracking, demand forecasting, complete accounts and a dedicated accountant/finance workspace for small sellers.

Document	Initial Business Idea Report (Revised)
Group members	Muhammad Anas, Muhammad Huzaifa
Date	31 October 2026
Revision	Version 2.1 - Accountant and finance module added

1. Formal Business Overview

StockPilot is a cloud inventory, demand-forecasting and accounts assistant for small retailers and online sellers. It tracks stock by product, predicts how much will sell, and recommends what to reorder and when. A new accounts module records sales, purchases, expenses, customer receivables and supplier payables in double entry, so the owner can see profit, cash and balances that match the stock records. A dedicated accountant and finance workspace then turns those records into active control: budgets, cash-flow forecasts, reconciliation queues, month-end close checklists and accountant sign-off. Alerts and a daily summary go to the owner on WhatsApp. It is designed to be set up from an Excel file in minutes and to work alongside the sales tools sellers already use.

Target segment	Examples	Why they fit
Online sellers (primary)	Shopify, WooCommerce and Instagram brands in clothing, cosmetics, electronics accessories	Many variants, fast-moving trends, cash tied up in stock
Small retail shops	General stores, garment shops and accessory shops with 100 to 5,000 products	Stock counted by hand or in Excel; frequent stockouts
Small distributors	Wholesalers supplying shops	Large purchasing decisions and supplier credit

2. Market Problem

Small sellers lose sales when popular items run out and lose cash when slow items pile up. Most keep stock records in Excel or notebooks that are updated late and say nothing about what to buy next. Their financial records are usually separate, incomplete, or prepared only when the accountant asks for them, and no one actively tracks budgets, cash-flow gaps or a clean month-end close.

- **Stockouts and overstock are costly worldwide.** IHL Group research reports about $1.7 trillion lost in 2026, or 6.2% of worldwide retail sales, to inventory distortion (overstocks and out-of-stocks), according to Retail Insight Network.
- **Small operators feel it often.** A 2026 survey of 400 inventory operators by inFlow found 44% stock out monthly, 54% pay 10% or more to hold stock, and 80% lose 1% to 5% of stock a year to shrinkage. The survey is not limited to small businesses.
- **Local pain points match.** A Pakistani retail software guide lists tracking many products, dead stock, reordering before stockouts and seasonal swings as common problems, and says Excel is slow and updates stock only at the end of the day (HysabOne, a vendor article).
- **Books do not match stock.** Sales, purchases and expenses sit in different places, so profit, receivables, payables and stock value rarely agree. Owners cannot tell which items make money or how much cash is tied up.

Competition

Alternative	What it offers	Gap StockPilot can use
HysabOne	Cloud POS, inventory, customer credit (khata) and tax invoices for Pakistani shops; vendor article estimates software at PKR 2,000 to 10,000 per month	Counter-billing focus; no stated demand forecasting or suggested purchase orders
Oscar POS	Cloud POS and inventory for retail and restaurants with low-stock alerts	Billing first; forecasting and purchasing advice not described
Loyverse POS	Basic retail billing and inventory	Advanced reporting may need extra integrations
Accounting software (QuickBooks, Xero, local packages)	Double-entry books, invoicing, tax reports and financial statements	Not inventory-aware in the StockPilot sense; separate data entry; no reorder or dead-stock decisions
Excel and registers	Free and familiar	Late, error-prone, no forecasts and no reliable books

POS tools are strong at billing. Accounting tools are strong at records. The gap is the combined decision layer: how much to order, from whom and by when, and how that decision affects profit, stock value and cash. Local market size has not been verified and will be tested in seller interviews.

3. Proposed Solution and What Makes It Unique

Module	What it does
1. Fast setup	Upload an Excel or CSV file, or connect Shopify or WooCommerce; products and opening stock appear in minutes
2. Live stock ledger	Stock by product, variant and location, updated by sales, purchases and returns; barcode scanning from a phone camera
3. Reorder alerts	WhatsApp and email alerts when an item will run out before the next supplier delivery
4. Demand forecasting	Seasonal sales forecasts per item, with plain-language reasons such as "sells more before Eid"
5. Suggested purchase orders	A ready purchase list per supplier, with quantities, expected cost and supplier balance
6. Profit and dead-stock report	Margin by item and a list of items that have not sold, so cash is not stuck
7. Accounts and books (new)	Double-entry ledger for sales, purchases, payments, expenses and stock; customer and supplier balances; trial balance, profit and loss, balance sheet and cash reports
8. Accountant and finance workspace (new)	Accountant login and role-based access, review queue, bank/cash reconciliation, budgets, cash-flow forecast, month-end close checklist and sign-off, tax pack and a client switcher for accountants managing several shops

Unique points

- **Decisions, not just records:** StockPilot says "order 40 units by Friday" and explains why.
- **Works alongside existing tools** rather than forcing a new POS.
- **WhatsApp-first:** owners get alerts and a daily summary where they already spend their day.
- **Priced for small sellers** at the low end of local software price ranges, with a 14-day trial.
- **Connects to Munshi:** orders confirmed in chat can reduce stock automatically.

- **Books that match stock:** every sale, purchase and return posts to the ledger, so profit, stock value and cash can be reconciled instead of rebuilt in Excel.
- **Finance control, not just bookkeeping:** budgets, cash-flow forecasts, reconciliation queues and month-end close show what needs attention before the accountant asks.
- **Accountant-ready:** a dedicated workspace with role-based access, review and sign-off, a standard trial balance, profit and loss, balance sheet, receivables and payables aging, and a clean period close.

3.1 Accounts module (new)

The original StockPilot idea tracked stock and supplier balances but stopped short of a full accounting record. The accounts module closes that gap. It turns each operational event into a balanced journal entry, so the owner and the accountant work from the same numbers. The module is included in the subscription rather than sold as a separate product, because it increases retention and makes the inventory advice more credible.

Capability	What it provides
General ledger	Double-entry journal for sales, purchases, payments, expenses, returns and adjustments; immutable posted entries with reversal-only corrections and an audit trail
Sales and receivables	Sales invoices, customer receipts, credit notes, customer balances and receivables aging
Purchases and payables	Purchase bills, supplier payments, supplier balances and payables aging, linked to stock receipts
Cash and bank	Cash accounts, bank accounts, transfers and simple statement reconciliation
Expenses and owner	Expense categories, owner drawings and owner contributions
Inventory accounting	Inventory asset, cost of goods sold using moving weighted average, stock adjustments and sales returns
Tax	Configurable sales tax and input tax codes with effective dates; tax summary for the accountant
Reports	Trial balance, profit and loss, balance sheet, cash movement, general ledger, receivables and payables aging
Period close	Soft close, lock, year-end close and snapshots; reopening a locked period requires owner permission and an audit reason

> **Operating model.** The ledger is append-only and every entry must balance. Corrections are made by reversing the original entry, never by editing history. Money is stored in paisa, not floating point. The base currency is PKR for v1. Stock is valued at moving weighted average, and inventory plus accounting post in the same transaction so the two reports cannot drift apart silently.

3.2 Accountant and finance module (new)

The accounts module records what happened; the accountant and finance module helps control what happens next. It is a role-based workspace over the same ledger, not a separate accounting product, so the owner and external accountant always review the same numbers. Core finance tools such as budgets, cash-flow forecasts and reconciliation are available from the Plus plan; the full accountant workspace, approvals and multi-client switching are part of Business and the later Partner plan.

Capability	What it provides
Finance dashboard	Cash position, profit trend, receivables and payables, stock value, budget versus actual and upcoming cash gaps in one view
Accountant access and controls	Separate accountant login, role-based permissions, read-only reviewer access, approval steps and an audit trail; posted entries cannot be edited
Review queue	Unreconciled bank or cash items, missing bills, negative stock, tax exceptions, unmatched payments and balances that fail control checks
Reconciliation workspace	Import bank or cash statements, match and unmatch transactions, record differences and prove inventory, receivable and payable control balances
Budgets and cash-flow forecast	Monthly budgets by category, a rolling cash-flow forecast from sales, purchases, receivables and payables, and alerts before a shortfall
Month-end close and sign-off	Close checklist, open-item review, trial balance review, accountant sign-off, period lock and a required reason when a locked period is reopened
Receivables and payables follow-up	Aging views, reminder drafts, promise-to-pay notes and supplier payment planning so cash collection is managed, not only recorded
Tax and hand-off pack	Sales tax and input tax summaries, general ledger extract, financial statements and an Excel or PDF pack for the accountant; statutory filing remains outside v1
Partner workspace	Client switcher for accountants managing several shops, with per-client isolation, consent and a consistent review process
Documents and collaboration	Attach invoices and bills, add comments or queries, track sign-off and keep the evidence behind each adjustment with the entry

> **Module boundary.** The accountant and finance module is a workflow and control layer over the ledger. It does not replace the accountant's professional judgment or provide an audit opinion. Payroll, fixed assets and depreciation, full statutory tax filing, multi-currency, consolidation and automated bank feeds remain out of scope for v1.

Included in the subscription

The core ledger, cash and bank, expenses, customer and supplier balances, trial balance, profit and loss and balance sheet are available to all paying plans. Basic includes a simple finance dashboard; Plus adds budgets, cash-flow forecasting, reconciliation and month-end close; Business adds the full accountant workspace, approvals and partner client switching.

Deliberately out of scope for v1

Payroll, fixed assets and depreciation, multi-currency, consolidation, full statutory tax filing, automated bank feeds and a formal audit opinion are deferred. They add regulatory and operational complexity without proving the core value: reliable books and better cash decisions for a small seller.

The core engine - the posting pipeline, data model, controls and Python technology stack - is specified in the companion report, *StockPilot Accounts Module: Core Engine Technical Design Report*.

4. Revenue Model

Plan	Price per month (PKR)	Limits and features	Target customer
Basic	2,999	1 location, up to 500 products, stock ledger, low-stock alerts, core ledger, cash and bank, expenses, trial balance, simple finance dashboard	New online sellers, small shops
Plus	5,999	3 locations, up to 2,500 products, forecasting, purchase orders, supplier ledger, receivables and payables, profit and loss, balance sheet, budgets, cash-flow forecast, reconciliation, month-end close	Growing brands and shops
Business	11,999	10 locations, up to 10,000 products, multiple users, API access, priority support, full accountant workspace, approvals, partner client switching, multi-location reports	Distributors and chains

Income sources

1. **Monthly subscriptions** (main source), with a discount for annual prepayment.
2. **Onboarding service:** PKR 5,000 one time for data cleaning, opening balances, setup help and accountant handover, including the accounts and finance module.
3. **Integration add-ons** for additional stores, ERP or accounting tools.
4. **Partner plan** for accountants and consultants managing several clients (phase 2).

Pricing strategy. Keep the entry price within the local range for small-business software, tie upgrades to features that save money (forecasts, purchase orders, finance control and accountant collaboration), and offer a 14-day trial. The accounts module is included in every plan rather than sold separately: it raises perceived value, supports retention and gives Basic customers a reason to upgrade when they need receivables, payables, budgets, reconciliation and formal statements. The advanced accountant workspace is a Business and Partner-plan feature because it carries more users, controls and support.

First-year projection (planning assumptions)

Item	Assumption or result
Paying customers by month 12	120, growing in a straight line from zero
Customer mix at month 12	70 Basic, 40 Plus, 10 Business
Monthly recurring revenue at month 12	PKR 569,880 (about PKR 6.8 million annualised)
Average revenue per customer	PKR 4,749 per month
Subscription revenue in year 1	About PKR 3.42 million
Onboarding fees in year 1	PKR 0.30 million (half of customers buy onboarding)
Total year-1 revenue	About PKR 3.72 million
Variable costs (hosting, messaging, payment fees)	Assumed 15% of revenue
Fixed monthly costs (tools, support, marketing)	Assumed PKR 150,000
Break-even	About 38 paying customers (150,000 divided by about 4,037 contribution per customer)

The projection does not add a separate accounting line because the accounts and accountant/finance modules are bundled into the existing plans. They are treated as retention, differentiation and upsell features; any extra onboarding and support effort is assumed to sit inside the existing fixed-cost allowance.

StockPilot has lower revenue per customer than Munshi and ERP Pilot, so it needs more customers. Its advantage is low running cost, self-serve onboarding and the chance to cross-sell to Munshi customers. The

accounts and accountant/finance modules add a second reason to stay: the owner's stock, profit, receivables and payables finally agree, and the accountant has a clean, shared workspace for review and close.

5. Technology Stack (Python)

The accounts and accountant/finance modules use the same Python stack as the rest of StockPilot. That keeps one deployable application, one database transaction for stock and accounting, and one team vocabulary. A separate accounting service would add distributed transactions and operational overhead that a small-seller product does not need.

Layer	Choice	Role
Language and backend	Python 3.12 with FastAPI	API, stock logic, accounting posting and webhooks
Database	PostgreSQL; SQLAlchemy and Alembic	Products, stock movements, suppliers, sales history, journal, ledgers and periods
Background jobs	Celery with Redis	Nightly forecast runs, alerts, daily summaries, report exports and reconciliation
Forecasting	pandas, statsmodels (exponential smoothing) and scikit-learn	Start with a seasonal baseline; compare models by backtesting on each customer's own sales history
Accounts engine	Python double-entry posting rules; append-only journal; PostgreSQL constraints; SQLAlchemy	Sales, purchases, payments, expenses, trial balance, profit and loss, balance sheet and period close
Finance workspace	FastAPI and HTMX dashboards; Celery schedules; openpyxl exports	Budget versus actual, cash-flow forecast, reconciliation queue, close checklist, accountant review and hand-off pack
Data import and export	openpyxl and pandas	Excel and CSV import, opening balances, reports and exports
Interface	FastAPI with Jinja2 and HTMX; Plotly charts; browser barcode scanning	Dashboard, phone-friendly stock counting and accounting screens
Messaging	WhatsApp Business Platform Cloud API via httpx	Alerts, daily summaries and receivable/payable reminders
Integrations	Shopify and WooCommerce APIs	Sync products, orders, stock levels and sales for posting
AI (optional)	LLM API through its Python SDK	Explain forecasts and answer questions such as "what is selling slowly?"
Operations	Docker, GitHub Actions, pytest, Caddy or Nginx, Sentry	Deployment, testing and monitoring
Accounting tests	pytest, Hypothesis and testcontainers	Balance invariants, reversals, idempotency, concurrency and reconciliation against real PostgreSQL

Forecast quality will be checked against simple rules such as a moving average, so a model is shipped only if it beats the baseline on real sales history. The accounts engine applies the same discipline: every posting rule has a test that proves the entry balances, and a nightly reconciliation checks that materialised balances, receivables, payables and inventory value still agree with the journal. The finance workspace reads from the same journal and control checks, so its dashboard and close checklist cannot silently use a second set of numbers.

6. Risks and Objective Summary

Risk	Response
Sellers do not keep clean sales history	Start with simple rules and improve forecasts as data grows; help with data cleaning at onboarding
Low price per customer	Self-serve onboarding, low support load, upsell to Plus and Munshi
POS vendors add forecasting	Stay integration-friendly and focus on purchasing decisions and WhatsApp-first use
Wrong forecast causes a bad purchase	Show reasons and ranges, keep the owner in control, track forecast accuracy
Hard to reach small sellers	Start with one niche and partner with accountants and suppliers
Accounting errors or wrong tax treatment	Double-entry invariants, reversal-only corrections, accountant review, pilot parallel run; tax codes are configurable and reviewed locally
Opening balances are messy	Staged Excel import with validation, preview and reconciliation; accountant-assisted onboarding for stores with existing books
Accountant does not adopt another portal	Read-only access, client switching, Excel and PDF hand-off packs and a short close checklist; the owner can still export everything
Budgets and cash forecasts are wrong	Show assumptions and actual-versus-plan, let the owner edit drivers, and use alerts as prompts rather than automatic decisions

Objective. This report shows that StockPilot addresses a real and costly gap (stock and financial decisions made by guesswork), that existing local tools focus on billing or accounting but rarely both with inventory, and that a low-priced subscription can reach break-even at about 38 customers, with first-year revenue of roughly PKR 3.7 million under stated assumptions. Seller interviews and a pilot with five stores will test these assumptions before the full product is built. The accounts and accountant/finance modules turn StockPilot from a stock tracker into a complete inventory, books and finance-control product for a small seller.

Sources

Figures were taken from web pages read on 18 October 2026. Prices, customer numbers, costs and break-even points are the group's own planning assumptions, not sourced data, and will be tested with real customers. Global inventory figures are not specific to small Pakistani businesses. The accounts and accountant/finance module design is an internal planning proposal, not a sourced market claim.

- Overstock, stockouts and returns cost retail, Retail Insight Network (IHL Group research): https://www.retail-insight-network.com/features/how-overstock-stockouts-and-returns-cost-retail-1-75tn/
- State of Inventory 2026, inFlow: https://www.inflowinventory.com/state-of-inventory-2026
- Best accounting software for retail shops in Pakistan, HysabOne (promotional): https://blog.hysabone.com/?p=17
- Best POS system in Pakistan, Oscar (promotional): https://oscar.pk/blog/best-pos-system-pakistan
- Small business automation statistics 2026, Stealth Agents: https://stealthagents.com/research/small-business-automation-statistics-2026
- WhatsApp Business gets AI agent, ProPakistani: https://propakistani.pk/2026/06/06/whatsapp-business-gets-ai-agent-to-handle-customer-chats/

Prepared for the StockPilot project. This is a business planning document, not a statutory accounting or tax opinion. Posting rules, tax codes and report definitions must be reviewed by a qualified accountant before production use.